Antarctic Blue Ice Classification Using Sentinel-1/2: An Application in the Lambert Glacier Basin

Yimeng Zhou, Lei Zheng, Fengming Hui, Rui Xu, and Xiao Cheng

***Abstract*— The Antarctic blue ice can be classified into wind- and melt-induced based on their origins. They play a different role in the development of surface water systems, the surface energy balance, and the infrastructure. Currently, visible light remote sensing is the most effective method for mapping blue ice. However, optical imagery faces difficulties in classifying blue ice accurately, and it is also greatly influenced by weather conditions. Synthetic aperture radar (SAR) images have the potential to map blue ice under all weather conditions, but it is difficult to distinguish blue ice from other similar weak microwave reflecting surfaces. In this study, by employing a segmentation method based on band ratios of the Sentinel-2 images, we delineated the geographical distribution of blue ice area (BIA) in the Lambert Glacier Basin (LGB). Taking advantage of the disparity in coherence levels between melt- and wind-induced blue ice, we performed blue ice classification in the LGB using Sentinel-1 images. The proposed method achieves an overall accuracy of 0.91 and $F1$-score of 0.91 and provides blue ice types with a spatial resolution of 10 m. The total area of blue ice in the Lambert Basin was estimated to be approximately 1.986×10^4 km^2. Among them, the area of melt-induced blue ice was approximately 1.276×10^4 km^2, while the wind-induced blue ice covered around 0.710×10^4 km^2. Melt-induced BIA was predominantly distributed in low-altitude coastal areas and downstream of glaciers, exhibiting higher surface temperatures compared to wind-induced BIA. Wind-induced BIA, on the other hand, was mainly found near nunataks and exposed rocks, displaying higher albedo than melt-induced BIA.**

***Index Terms*— Antarctica, blue ice, coherence, microwave.**

I. Introduction

BLUE ice is an ancient ice with a flat and hard surface, emerging after the removal of the firn layer [1]. Despite covering only approximately 1% of the Antarctic [2], blue ice areas (BIAs) have long been a focus of Antarctic investigations. It has practical applications in meteorite searches, paleoclimate studies, and Antarctic airport construction [3], [4], [5]. Winther et al. [2] divided blue ice into two categories, namely, wind-induced blue ice and melt-induced blue ice. Wind-induced BIA predominantly undergoes wind erosion and sublimation and is often found near mountain ranges or areas with significant topographic variations [1]. Melt-induced BIA experiences surface and near-surface melting and is primarily distributed in coastal regions. Unlike the traditional definition of blue ice, melt-induced blue ice does not originate from glaciers but rather from cyclic melt-freeze processes [2], [6]. By investigating the temporal changes in the extent and characteristics of surface melt in melt-induced BIAs, it is able to assess the sensitivity of BIAs to atmospheric forcing [7], [8], [9], [10], thus monitoring changes in Antarctic climate and the variability in the energy budget and mass balance of the ice sheet [1], [11], [12], [13], [14].

Given the unique geographical location and extreme climatic conditions of Antarctica, research in BIAs predominantly relies on satellite remote sensing techniques. To date, numerous studies have been conducted to monitor and map BIAs using optical imagery [14], [15], [16]. By exploiting the spectral distinctions between BIAs and surrounding land features, such as snow and bare rocks, the band ratio approach has demonstrated its effectiveness in distinguishing BIAs [2], [17], [18], [19], [20], [21]. The earliest all-Antarctic BIAs map was obtained using an image mosaic from the U.S. National Oceanic and Atmospheric Administration (NOAA) Advanced Very-High Resolution Radiometer (AVHRR) sensor [2]. Subsequently, Hui et al. [18] used the Landsat Image Mosaic of Antarctica (LIMA) [22] to map Antarctica BIAs at a higher resolution. Although Wei et al. [20] have obtained an updated map of blue ice coverage based on Landsat 8 imagery, it is necessary to obtain a more current and precise map of blue ice coverage due to potential changes that may have occurred in recent years.

In addition to the band ratio method, studies have been conducted to delineate the boundaries of BIAs based on physical parameters, such as snow grain size and surface albedo. Scambos et al. [15] used the Mosaic of Antarctica snow grain size product (MOA2003/2004) to successfully map the extent of blue ice throughout Antarctica. Luis et al. [16] found a notable correlation between albedo and BIAs obtained from Landsat 8 imagery over the Polar Record Glacier, and an albedo threshold was used to effectively delineate blue ice extent.

Early studies utilized spectral and texture features to differentiate various types of BIA [2], [23]. The presence of snow crystals and air bubbles within the ice can lead to an increase in its reflectivity. In regions of low elevation, the ice-bound snow crystal content in melt-induced BIAs is often higher due to the repeated freeze-thaw cycles, compared

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to wind-induced BIAs [2], [8], [41]. Nonetheless, the presence of liquid water somewhat diminishes this distinction. Recently, Hu et al. [6] developed a daily product of fractional austral-summer blue ice over Antarctica (FABIAN) based on Moderate-Resolution Imaging Spectroradiometer (MODIS) imagery and machine learning methods. FABIAN can monitor interannual and intra-annual variations in BIAs and distinguish between different BIA types based on the degree of change. However, the mapping resolution from MODIS is relatively low. Moreover, machine learning relies on a large volume of data for lengthy training periods, and the accuracy of the results is heavily influenced by the quality of the training samples.

Apart from optical imagery, studies have been carried out to map BIAs based on synthetic aperture radar (SAR) images [24], [25]. Blue ice, characterized by its low primarily specular reflection, exhibits a backscatter coefficient that falls between that of wet snow and dry snow. Liu et al. [24] differentiated blue ice, dry/wet snow, and the firn line by defining a threshold for the backscatter coefficient. Due to its stable nature, blue ice demonstrates high coherence. Cheng et al. [25] utilized ERS-1/2 data to obtain interferometric coherence images for mapping the Grove Mountain BIAs. However, these methods tend to exhibit lower accuracy compared to optical approaches. Despite the advantages of SAR imagery, such as its ability to overcome limitations posed by light, weather, and clouds, there is currently no high-precision SAR-based method for BIA extraction and classification. The received radar backscatter from melt- and wind-induced blue ice is expected to be different due to the difference in surface/subsurface structure. Nevertheless, at present, it is still challenging to distinguish blue ice types using SAR images as the radar backscatter is always noisy, and the interferometric coherence can be easily interfered by the changes in firn/snow features.

This study aims to develop an efficient and rapid blue ice mapping and classification method based on Sentinel-1/2 images. Compared with the coarse-resolution MODIS and high-noisy SAR images, the high-resolution satellite images from Sentinel-2 provide a way to monitor the detailed distribution patterns of Antarctic BIA. The blue ice types within the derived BIAs are afterward determined by the interferometric coherence derived from Sentinel-1 images. The Lambert Glacier Basin (LGB) was selected as the study area as it encompasses a substantial number of both melt-induced BIAs and wind-induced BIAs. The spatial distribution of two types of blue ice over the LGB and their effects on surface energy balance are further investigated.

II. Study Area and Datasets

A. Lambert Glacier Basin

The LGB is located between 68.5°–81°S and 40°–95°E and is composed of the Lambert, Mellor, Fisher glaciers, and several glacier tributaries (Fig. 1) [26]. It covers a total area of approximately 1.55×10^6 km^2, including 69 000 km^2 of floating ice in the Amery Ice Shelf [27]. A large number of BIAs are distributed within the LGB, accounting for approximately 23% of the total BIAs in Antarctica [2]. As the largest gathering place of blue ice and one of the largest glacier basins in Antarctica, the LGB holds significant importance in the mass balance of the entire Antarctic ice sheet.

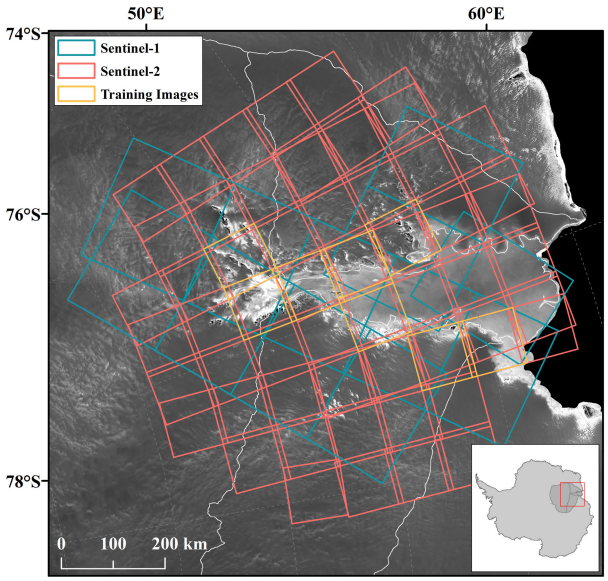


Fig. 1. LGB and its location in Antarctica (dark gray area in the quick look). Blue, pink, and yellow framed lines are the frames of Sentinel-1/2 and training images used in this study. The red box in the quick look indicates the region shown in the major figure.

B. Sentinel-2

Sentinel-2A&B is a constellation of two satellites that provides a revisit time of five days. It is equipped with a multispectral imager (MSI) that captures 13 spectral bands with a spatial resolution ranging from 10 to 60 m. The optical imagery obtained by Sentinel-2 undergoes rigorous preprocessing steps, including orthorectification and geometric correction, to produce Level-1C (L1C) products.

In this study, we leveraged the Google Earth Engine (GEE) platform to acquire cloud-free or low-cloud Sentinel-2 optical remote sensing products L1C for the summer seasons of 2020/2021 to 2021/2022 in the LGB. The main objective of this study was to delineate BIA using the acquired Sentinel-2 imagery. To address the issue of cloud cover in the optical imagery, we adopted the C Function of Mask (CFMask) algorithm embedded in GEE and utilized the quality assessment (QA) band QA for cloud removal. The data used in this study were obtained from the Copernicus Data Hub of the European Space Agency (ESA, https://scihub.copernicus.eu).

C. Sentinel-1

Sentinel-1A&B is a polar-orbiting satellite constellation comprising two satellites that are situated in the same orbital plane and equipped with a C-band SAR sensor. In this study, we obtained 24 images of Sentinel-1 Interferometric Wide swath (IW) Single Look Complex (SLC) SAR in HH polarization covering the LGB from January 13 to January 30, 2021. The spatial resolution of the imagery is 5×20 m, and the detailed information is presented in Table I. Orbital types A and D in the table represent the ascending and descending orbits, respectively.

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TABLE I
SENTINEL-1 SAR DATASET USED IN THIS STUDY

Acquisition Date	Path	Frame	Orbit Type	Acquisition Date	Path	Frame	Orbit Type
20210115	28	938	A	20210127	28	938	A
20210118	72	938	A	20210130	72	938	A
20210118	72	933	A	20210130	72	933	A
20210113	3	834	D	20210125	3	834	D
20210113	3	839	D	20210125	3	839	D
20210113	3	843	D	20210125	3	843	D
20210113	3	848	D	20210125	3	843	D
20210113	3	854	D	20210125	3	854	D
20210116	134	858	D	20210128	134	858	D
20210116	134	853	D	20210128	134	853	D
20210116	134	848	D	20210128	134	848	D
20210116	134	843	D	20210128	134	843	D

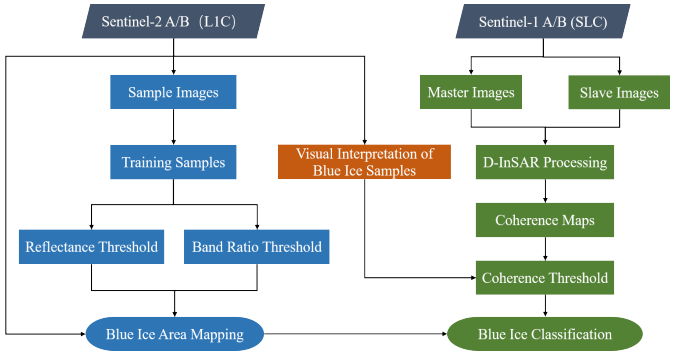


Fig. 2. Flowchart for blue ice extraction and classification based on Sentinel-1/2.

The dataset includes six ascending and 18 descending orbit images, forming 12 pairs of interferometric SAR (InSAR) images for generating interferograms and performing blue ice classification in the LGB. The data preprocessing and analysis methodology are described in detail in Section III-B. The data were obtained from the Alaska Satellite Facility (ASF) of the National Aeronautics and Space Administration (NASA).

III. METHODS

This section described the methodology employed in blue ice classification using Sentinel-1/2 images (Fig. 2). In Section III-A, we elucidate the process for blue ice extraction using Sentinel-1 optical images. Subsequently, in Section III-B, we introduce the underlying principles and influential factors associated with interferometric coherence and outline a specific approach for blue ice classification that relies on the analysis of interferometric coherence derived from Sentinel-1 SAR images.

A. BIA Extraction Based on Reflectance From Sentinel-2 Images

The band ratio method has been extensively employed for BIA extraction research owing to its superior performance. Its underlying principle is based on exploiting the spectral characteristics of BIA to differentiate it from other land cover types, such as snow and bare rocks [2], [17], [18], [19], [20], [21]. The normalized difference snow index (NDSI) was first applied to blue ice mapping by Brown and Scambos [17].

TABLE II
SENTINEL-2 IMAGES THAT EMPLOYED IN TRAINING AND VALIDATION

No.	ID of Sentinel-2 images	Acquisition date	Cloud cover (%)
1	S2B_MSIL1C_20201123T034629_N0209_R075_T42DWF_20201123T054934	2020.11.23	5.794
2	S2B_MSIL1C_20211228T034619_N0301_R075_T43DDB_20211228T053513	2021.12.28	1.652
3	S2B_MSIL1C_20220215T041719_N0400_R061_T42CVE_20220215T060100	2022.02.15	0.842
4	S2B_MSIL1C_20220126T041719_N0400_R061_T41CNU_20220126T055324	2022.01.26	3.406
5	S2B_MSIL1C_20211227T041719_N0301_R061_T41DPB_20211227T055212	2021.12.27	3.591
6	S2B_MSIL1C_20220216T034619_N0400_R075_T42CVD_20220216T064913	2022.02.16	0.244
7	S2B_MSIL1C_20210221T034629_N0209_R075_T43DDC_20210221T055036	2021.02.21	1.066
8	S2B_MSIL1C_20210131T041729_N0209_R061_T42DVF_20210131T072137	2021.01.31	2.574

It is defined as the ratio between the difference and the sum of the reflectance values in the green and near-infrared (NIR) bands. They demonstrated that NDSI can be effectively used for long-term monitoring of blue ice extent. Subsequent research built upon this concept and improved it by replacing the green band with the blue band, defining the normalized difference blue ice index (NDBI), which yielded better results for BIA mapping [20]. We conducted a set of experiments that employed band ratio methods for BIA detection [18]. The experiments indicated that the combination of the blue and NIR bands performed the best for BIA extraction. Therefore, the NDBI for BIA extraction was calculated using bands 2 and 8 from Sentinel-2 imagery in this study. The formula is as follows:

$$\mathrm{NDBI} = \frac{R_2 - R_8}{R_2 + R_8} \tag{1}$$

where R_2 and R_8 are the reflectance of blue and NIR bands, respectively. The optimal threshold, denoted as R_{BI}, for blue ice extraction, was determined by selecting the NDBI parameter that corresponds to the best results in terms of blue ice detection.

Optical imagery can be influenced by various factors, such as terrain conditions, atmospheric conditions during image acquisition, geographic location, and solar zenith angle. To improve the generalizability of R_{BI}, we selected eight sample images randomly from different locations and times within the LGB. All of the selected samples underwent rigorous preprocessing steps, including orthorectification and geometric correction. The detailed information of the sample images is presented in Table II. Notably, due to the variations in reflectance among different Sentinel-2 bands, we sharpened the lower resolution band into 10-m spatial resolution by image fusion technology to facilitate the subsequent band synthesis and statistical analysis.

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TABLE III
MEAN NDBI PARAMETER OF EACH SURFACE FEATURE TYPE IN EACH SAMPLE IMAGED FROM TRAINING SAMPLES

No.	Blue ice	Rock	Snow	Shadowed rock	Shadowed snow or blue ice	Supraglacial lake
1	0.267	-0.015	0.068	\	\	\
2	0.203	-0.081	0.062	0.531	\	0.886
3	0.237	-0.058	0.076	0.416	0.437	0.419
4	0.194	-0.027	0.051	0.381	0.369	\
5	0.219	0.026	0.058	0.526	\	0.731
6	0.242	-0.042	0.041	0.389	0.442	0.410
7	0.187	-0.069	0.049	0.588	0.592	\
8	0.209	-0.038	0.064	0.524	\	0.453
Mean±standard deviation	0.219±0.025	-0.038±0.031	0.058±0.011	0.479±0.076	0.460±0.081	0.579±0.193

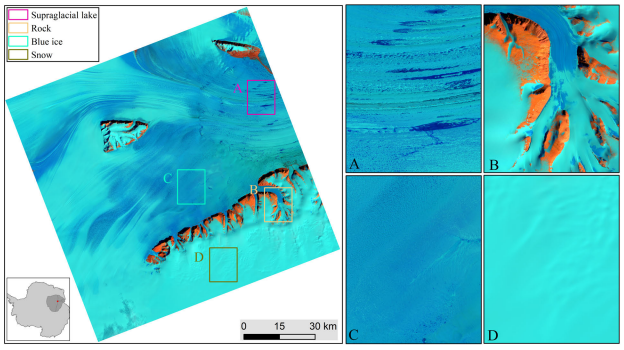


Fig. 3. Surface features on Sentinel-2 image (February 2021), A, B, C, and D show the regions with supraglacial lakes, rocks, blue ice, and snow, respectively.

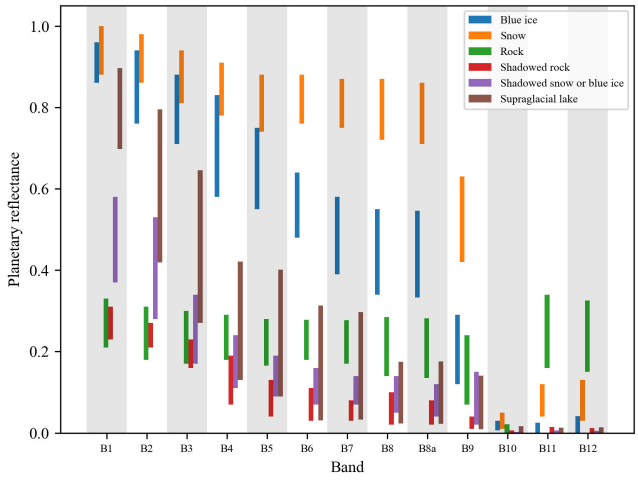


Fig. 4. Reflectance range for different ice sheet surface features.

This study utilized the BIA extraction method, as proposed by Hui et al. [18], to categorize the surface of LGB into six distinct types, including blue ice, rock, snow, shadowed rock, shadowed snow or blue ice, and supraglacial lake. Eight randomly selected sample images were analyzed by creating false-color composite images using bands 12, 8, and 2 for the training samples (Fig. 3). Visual interpretation was employed to select training samples for each land cover type. Subsequently, the NDBI parameter was computed for different land cover types within the sample images to determine R_{BI} for BIA extraction. It is important to note that the sampled areas displayed significant variability in land cover types. Thus, not all samples included all six land cover types mentioned above. Table III provides the statistical results.

Based on the results presented in Table III, the minimum NDBI parameter for blue ice among the eight sample images was 0.187. It was significantly higher than the NDBI parameters for rock (maximum value 0.026) and snow (maximum value 0.076). The NDBI parameter of shadowed rock and shadowed snow or blue ice exhibited substantial fluctuations, with the maximum reaching 0.588 and 0.592, respectively. The NDBI parameter of supraglacial lake samples collected in late December (0.7–0.9) was notably higher than those obtained from samples in the latter half of January (0.4–0.5), which are closely related to seasonal effects. Therefore, the criterion for identifying BIAs was set as $R_{\mathrm{BI}} > 0.144$. The threshold is determined by subtracting three times the standard deviation (SD) from the average.

Although the NDBI parameter proves effective in distinguishing between blue ice, rock, and snow, relying solely on R_{BI} alone is insufficient to differentiate blue ice from shadowed rock, shadowed snow or blue ice, as well as the supraglacial lake. We integrate both R_{BI} and a single-band threshold in the mapping of blue ice. To determine the optimal band and reflectance threshold for differentiating between blue ice and others, we computed the mean reflectance, SD, maximum, and minimum values of training samples for each of the six land cover types in different bands of sample imagery. The statistical results of band reflectance are presented in Table IV and Fig. 4.

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TABLE IV
STATISTICS (MEAN ± SD) OF REFLECTANCE OF ICE SHEET SURFACE FEATURES FROM TRAINING SAMPLES (THESE BANDS ARE THE SENSOR OF SENTINEL-2)

Band	Blue ice	Rock	Snow	Shadowed rock	Shadowed snow or blue ice	Supraglacial lake
1	0.904±0.028	0.291±0.057	0.937±0.043	0.301±0.044	0.420±0.070	0.769±0.041
2	0.878±0.029	0.263±0.059	0.911±0.048	0.260±0.049	0.395±0.058	0.667±0.055
3	0.781±0.031	0.247±0.059	0.857±0.037	0.196±0.056	0.251±0.029	0.482±0.061
4	0.719±0.021	0.260±0.040	0.847±0.043	0.136±0.038	0.159±0.052	0.257±0.086
5	0.687±0.021	0.264±0.041	0.838±0.038	0.110±0.032	0.139±0.040	0.218±0.089
6	0.608±0.026	0.255±0.063	0.840±0.040	0.096±0.023	0.121±0.028	0.164±0.087
7	0.545±0.030	0.263±0.034	0.830±0.031	0.088±0.017	0.114±0.021	0.143±0.073
8	0.503±0.045	0.254±0.029	0.798±0.023	0.069±0.016	0.108±0.026	0.111±0.057
8A	0.490±0.051	0.254±0.029	0.806±0.029	0.068±0.018	0.099±0.021	0.110±0.062
9	0.229±0.045	0.157±0.048	0.483±0.070	0.023±0.009	0.082±0.043	0.051±0.028
10	0.015±0.001	0.004±0.051	0.018±0.003	0.003±0.001	0.001±0.001	0.001±0.001
11	0.016±0.003	0.267±0.018	0.071±0.012	0.012±0.004	0.004±0.001	0.005±0.004
12	0.014±0.003	0.242±0.023	0.073±0.013	0.010±0.006	0.002±0.001	0.004±0.003
Number of pixels	1 572 739	328 016	2 874 593	86 750	51 421	36 270

The statistical results presented in Table IV demonstrate that blue ice exhibits a significantly higher average reflectance in band 8 (0.503) than shadowed snow or blue ice (0.108), shadowed rock (0.069), and supraglacial lake (0.111). In addition, as depicted in Fig. 4, the minimum reflectance of blue ice in band 8 is considerably higher than those of the other three surface features. Specifically, the NIR reflectance of blue ice ranges from 0.3 to 0.7, while the NIR reflectance of the supraglacial lake, shadowed snow or blue, and shadowed rock are below 0.2. Therefore, the regions satisfying the criteria of $R_{\text{blueice}} > 0.144$ and band 8 reflectance between 0.3 and 0.7 have been identified as BIAs.

B. Blue Ice Classification Based on Radar Interferometric Coherence From Sentinel-1 Images

In total, 12 pairs of InSAR images taken in January 2021 were employed in blue ice classification. With a temporal interval of 12 days, the Sentinel-1 image achieved a cover of the extent of BIAs in the LGB (Table I). Interferometric processing was conducted using the Environment for Visualizing Images SARscape software. Interferometric coherence images were generated for the purpose of blue ice classification.

Before generating coherence maps through interferometric processing, registration of the InSAR pairs is necessary. Each InSAR pair is processed with the second image, acquired in late January, serving as the master SLC image, and the first image, acquired in mid-January, serving as the slave SLC image after registration. It is a fundamental indicator for assessing the quality and coherence of two co-registered SAR images. The coherence coefficient quantifies the similarity between the two interferometric SAR images and the changes in the scattering characteristics of the terrain [28]. It is defined as

$$\gamma = \frac{E\left\langle T_1 T_2^* \right\rangle}{\sqrt{\left\langle \left| E^2(T_1) \right| \right\rangle \left\langle \left| E^2(T_2) \right| \right\rangle}} \tag{2}$$

where T_1 and T_2 denote the master and resampled slave image, respectively, and * represents complex conjugate multiplication.

In the actual calculation, it is generally replaced by the sample-averaged maximum likelihood coherence estimate [29]

$$\gamma = \frac{\left| \sum_{i=1}^{N} T_{1i} T_{2i}^* \right|}{\sqrt{\sum_{i=1}^{N} |T_{1i}|^2 \cdot \sum_{i=1}^{N} |T_{2i}|^2}}. \tag{3}$$

Radar interferometry includes various modes, such as along-track interferometry, cross-track interferometry, and repeat-pass interferometry. In this study, we used the most common mode of repeat-pass interferometry, where the interferometric phase can be expressed as the product of several phase components as follows [30]:

$$\gamma = \gamma_{\text{spatial}} \cdot \gamma_{\text{tempo}} \cdot \gamma_{\text{dop}} \cdot \gamma_{\text{snr}} \tag{4}$$

where γ_{spatial} and γ_{tempo} denote spatial decorrelation and temporal decorrelation, respectively, and γ_{dop} and γ_{snr} indicate the decorrelation caused by Doppler centroid discrepancy and thermal noise, respectively.

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TABLE V
MEAN AND SD OF THE COHERENCE FOR TWO TYPES OF BIA SAMPLES

	Melt-induced BIA	Wind-induced BIA
Mean	0.096	0.441
SD	0.051	0.119
Upper bound	0.198	\
Lower bound	\	0.203
Number of pixels	781683	784587

The original interferogram is generated by pixel-by-pixel complex conjugate multiplication between the master image and the registered slave image. Due to the spatial geometry of InSAR, there could be phase offsets that should be constant but change, and these offsets are removed by using the Reference Elevation Model of Antarctica (REMA) Mosaic Digital Elevation Model (DEM) to remove the flat Earth phase [40]. After flat Earth removal, the resulting interferogram is filtered using a Boxcar filter to suppress the interferometric noise [31]. In data processing, the azimuth and range multilook factors are set to 5 and 1, respectively. Both multilooking and filtering processing can improve the SNR of the interferogram and reduce the impact of thermal noise. The impact of geometric decorrelation can be neglected in our study due to the small perpendicular baseline of Sentinel-1 [32]. The time decorrelation occurs due to physical changes in the surface of the target area between two radar acquisitions [33]. Different types of land cover exhibit distinct temporal decorrelation characteristics, which forms the basis for using coherence as a feature for blue ice classification.

Due to the absence of an absolute reference for blue ice classification results, we constructed a sample set through visual interpretation based on differences in surface morphology between wind-induced BIA and melt-induced BIA, combined with Sentinel-2 optical images. In this process, we applied the nearest neighbor interpolation method to resample the coherence images, ensuring their alignment with the Sentinel-2 imagery. Due to intense wind scouring, the surface texture of wind-induced BIAs is usually smooth (Fig. 5). In summer, melt-induced BIA exhibits the gullies and cracks formed by meltwater. With the repeated freezing and thawing of snow accumulated in the cracks, there can be mixed pixels containing a combination of blue ice and snow. We calculated the coherence of the two types of BIA samples in the Lambert basin obtained from the Sentinel-1 dataset interferogram. The statistical results are shown in Fig. 5 and Table V.

According to Table V, there is a significant difference in coherence between the melt-induced BIA and wind-induced BIA during the austral summer. We define coherence $>$ 0.20 as wind-induced BIA (with a lower bound of 0.203) and coherence $<$ 0.20 as melt-induced BIA (with an upper bound of 0.198). It is important to note that the coherence level of BIA is closely related to the season, and the coherence threshold proposed in this study is only applicable to the study area and period.

In theory, the backscatter coefficient can also be utilized for the classification of blue ice. However, the statistical analysis of the backscatter coefficient from various BIA samples generated in this study indicates suboptimal effectiveness (Fig. 6).

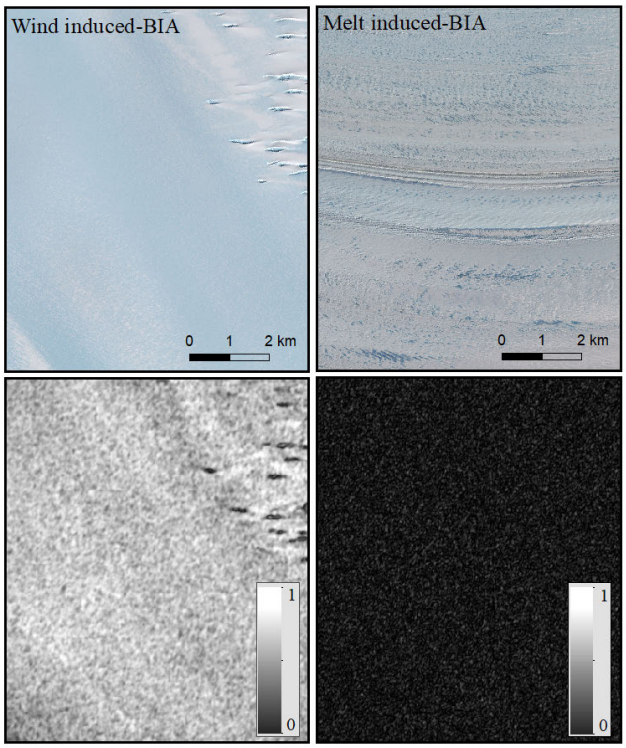


Fig. 5. Comparison between wind-induced BIA and melt-induced BIA on optical images and interferometric coherence images.

The wind-induced BIA, which undergoes repeated freezing and thawing, exhibits a much rougher surface compared to wind-induced BIA formed by katabatic wind action. Consequently, the backscatter coefficient of melt-induced blue ice is relatively higher. However, the presence of liquid water resulting from seasonal melting on the surface of melt-induced BIA can absorb electromagnetic wave signals, leading to a decrease in the backscatter coefficient. Significant differences in backscatter coefficients can be observed between the windward-facing and leeward-facing sides of the same type of BIA located at nunataks, suggesting that strong wind patterns may alter the arrangement of blue ice particles. In summary, although a few BIAs may exhibit uniform values in SAR images (such as the Grove Mountains), in most cases, the backscatter is influenced by various external factors, making it impossible to rely solely on the backscatter coefficient for determining BIA types.

The upper and lower bounds refer to the upper and lower limits calculated based on the statistical distribution algorithm, respectively. The formula is expressed as follows:

$$\frac{\text{Upper}}{\text{Lower}}\text{bound} = \text{Average} \pm 2 * \text{SD}. \tag{5}$$

C. Accuracy Assessment

In this study, we systematically chose validation samples distributed randomly within the training images through visual discrimination. These samples consisted of 65 231 blue ice pixels and 68 771 nonblue ice pixels. The acquired blue ice extent was juxtaposed with the validation samples to validate

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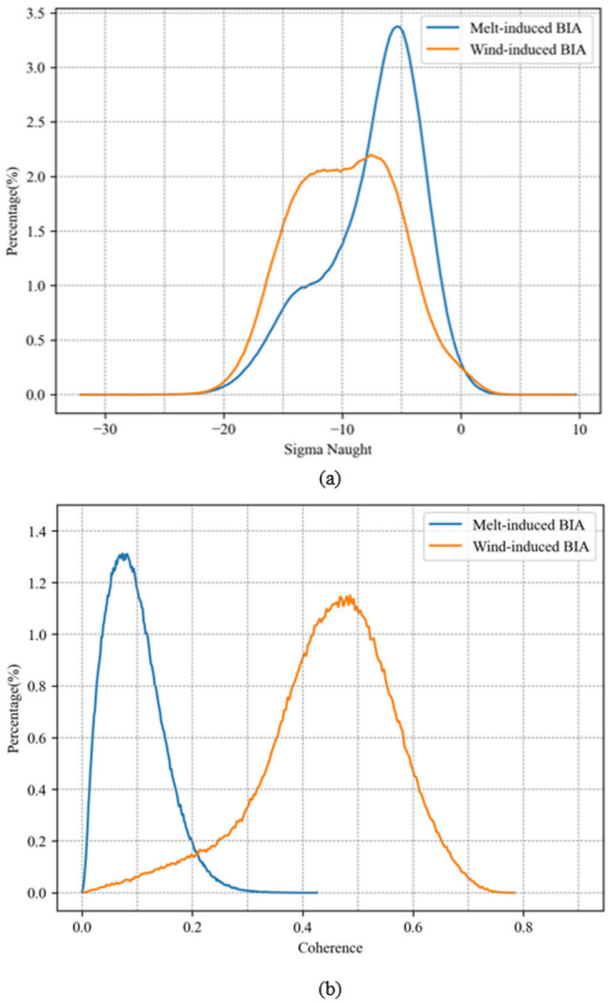


Fig. 6. (a) Backscatter coefficient of two types of BIA samples. (b) Coherence of two types of BIA samples.

the accuracy, with the $F1$-score serving as the measure of precision for BIA extraction.

The confusion matrix is employed in the validation with four types of results, namely, true positive (TP), false negative (FN), false positive (FP), and true negative (TN). $F1$-score was computed based on the precision rate and recall, ranging between 0 and 1. A higher $F1$-score value indicates a higher precision level. The calculation formula is as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{6}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

$$F1\text{score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}. \tag{8}$$

By statistical calculation, the $F1$-score for BIA extraction in this study was determined to be 0.939, with an overall accuracy of 0.940. While there were some instances where blue ice pixels covered by thin snow were not accurately extracted, the overall outcome was satisfactory, indicating that we have obtained a reliable BIA map of the LGB.

The same methodology was employed to assess the accuracy of the blue ice classification results. Considering the discrepancy in the area of the two types of BIA in the LGB, we randomly selected 37 271 validation sampling pixels in melt-induced BIAs and 27 565 validation sampling pixels in wind-induced BIAs.

The accuracy calculation results indicate that the $F1$-score for the melt-induced BIA is 0.922 and the $F1$-score for the wind-induced BIA is 0.891. The overall accuracy is determined by the ratio of correctly classified pixels to the total number of samples, which amounts to 0.909. Although there is still some confusion between the two types of BIA, overall, the results are satisfactory. It is indicated that the coherence-based blue ice classification method proposed in this study is feasible.

IV. Results

A. Distribution Patterns of Blue Ice

Through the above method, we generated a 10-m resolution BIA distribution map that includes both types of BIA (Fig. 7). For spatial characterization of BIAs, the BIA distribution map was overlaid with an automatically extracted rock outcrop dataset for Antarctica [34], which is from the Antarctic Digital Database (ADD). The interior area of the LGB is almost entirely covered with snow, and BIA is primarily distributed around the Amery Ice Shelf. The total area of BIA is 1.986×10^4 km^2, which represents only 1.28% of the total area of the LGB. The area of melt-induced BIA is approximately 1.276×10^4 km^2, while the area of wind-induced BIA is around 0.710×10^4 km^2. Fig. 7 illustrates the overall extent of BIA in the LGB from 2020 to 2022 and the specific distribution of the two types of BIA.

Fig. 7 indicates that melt-induced BIAs are primarily situated downstream of glaciers and upstream of ice shelves. Large accumulations of melt-induced BIA are formed at the confluence of Miller Glacier and Lambert Glacier with Amery Ice Shelf, as well as in the western Mawson Escarpment. In addition, a large area of melt-induced BIAs is found in the middle edge area of the Amery Ice Shelf, the southeastern part of Charybdis Glacier, and Gillock Island. Wind-induced BIAs are mainly found in mountainous areas such as Grove Mountains, Manzi Mountains, and the southern and eastern Prince Charles Mountains. In addition, wind-induced BIAs are also distributed in the upper Fisher Glacier, the eastern part of Miller Glacier, and Charybdis Glacier. A small amount of wind-induced BIA can also be found in the northeast side of Mawson Escarpment and Gillock Island.

To further investigate the spatial distribution characteristics of different types of BIA, we overlaid the obtained blue ice classification results of the LGB with the 10-m resolution REMA DEM products (Fig. 8), to determine the relationship between BIA distribution and elevation, as well as slope.

According to Fig. 8, the majority of the melt-induced BIA is located at elevations below 1000 m, accounting for about 93.39% of the total area of melt-induced BIA in LGB. Specifically, 64.74% of the area is below 500 m in elevation, 28.65% is between 500 and 1000 m in elevation, and 4.75% is between 1000 and 1500 m. Only 1.86% of the melt-induced BIA is located at elevations above 1500 m, while wind-induced BIA below 500 m accounts for 25.92% of the total wind-induced BIA in the LGB. In addition, 34.10% of the wind-induced BIA is in the area of 500–1000 m, 26.06% is above 1000–1500 m, and 13.92% of the wind-induced BIA is above 1500 m.

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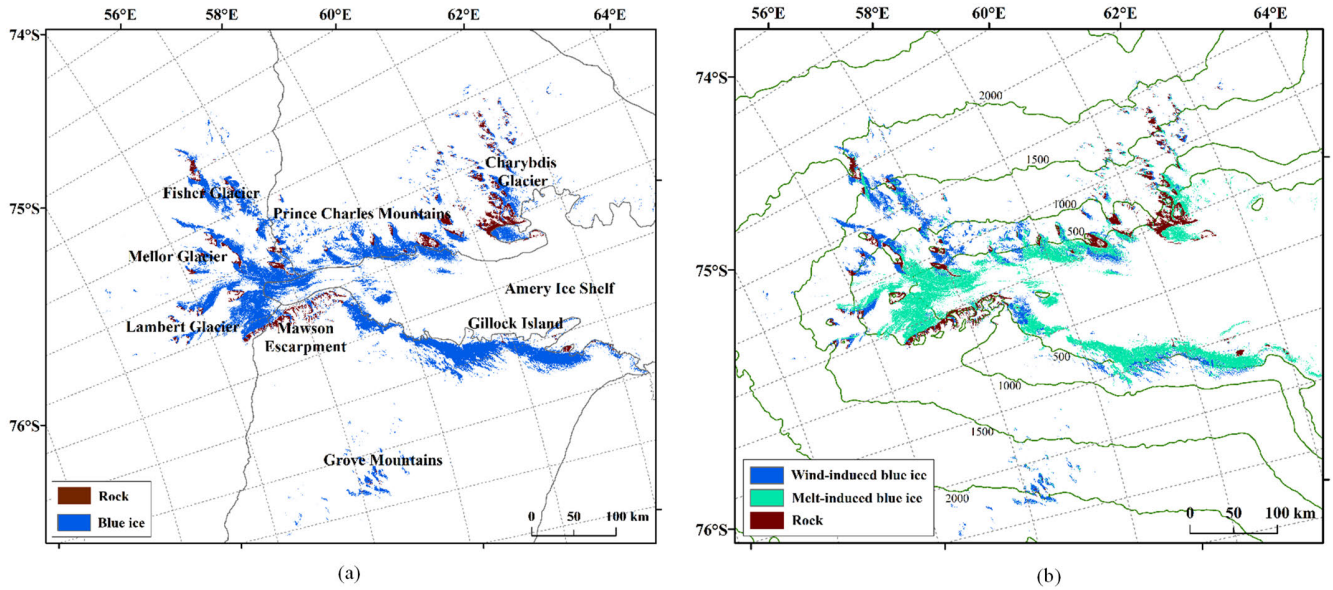


Fig. 7. Distribution of blue ice in the LGB. (a) Extent of blue ice from Sentinel-2 images. (b) Melt-induced blue ice and wind-induced blue ice from Sentinel-1 images. Green lines are elevation contours (m).

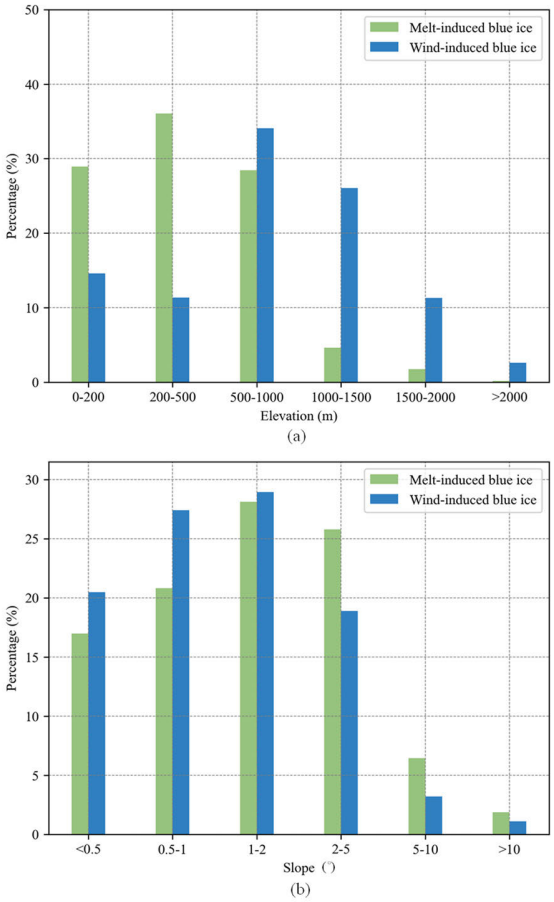


Fig. 8. Percentage of two types of BIA at different (a) elevations and (b) slopes.

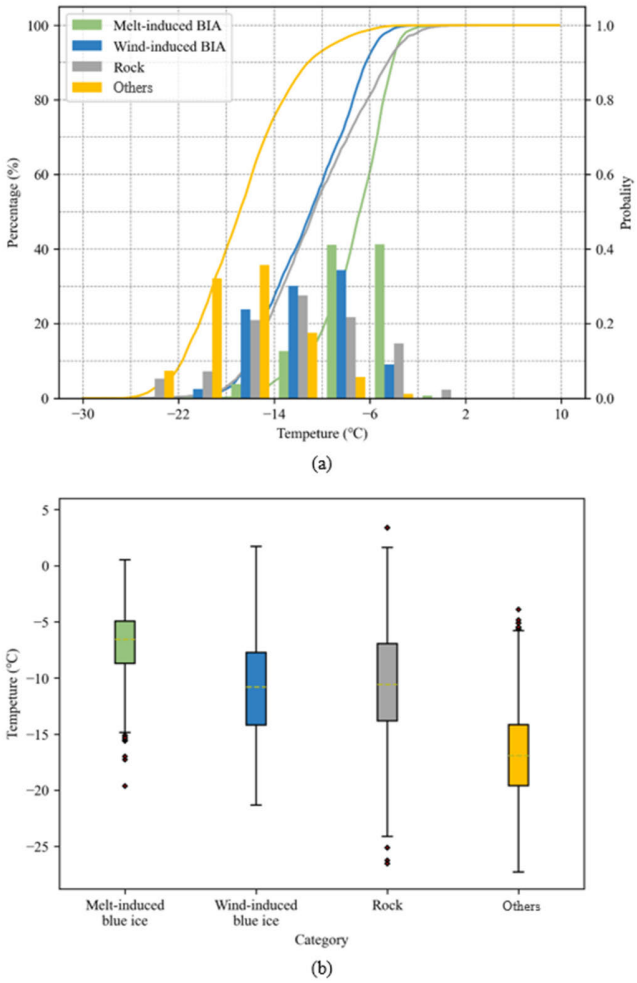


Fig. 9. (a) Probability distributions and (b) boxplot of temperature of different surface features.

The slope characteristics of the two types of BIA exhibit similarities, with only 1.98% of melt-induced BIA and 1.11% of wind-induced BIA occurring in the area with a slope greater than 10°. Melt-induced BIA and wind-induced BIA are primarily distributed in areas with slopes between 0° and 5°, accounting for a combined area share of 91.26% and 95.69%, respectively.

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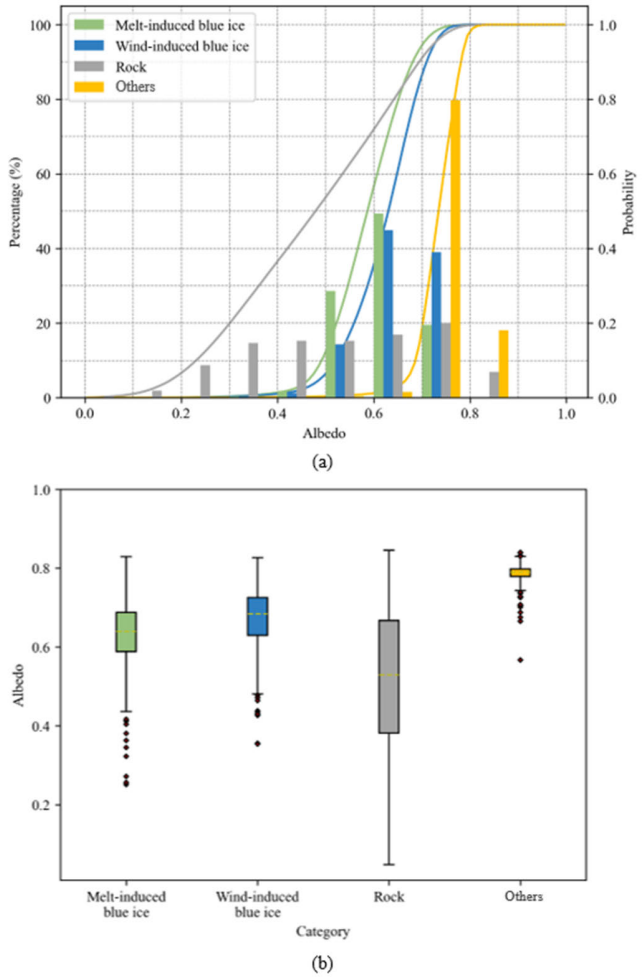


Fig. 10. (a) Probability distributions and (b) boxplot of albedo of different surface features.

B. Thermal Conditions of Blue Ice

To investigate the atmospheric characteristics of the two types of blue ice and compare them with other surface features, we divided ground cover into four categories: melt-induced blue ice, wind-induced blue ice, rock, and others. The "other" category is mostly comprised of snow, with a few additional features such as supraglacial lakes. Fig. 9 displays the temperatures of these four features during mid-January 2021. The MYD11A2 surface temperature product with a resolution of 1 km was utilized for this analysis. The mean temperatures of melt-induced BIAs are mainly falling between −14 °C and −4 °C, with an average of −7.2 °C. The surface temperature of wind-induced BIAs is generally lower than that of melt-induced BIAs, with temperatures uniformly distributed between −18 °C and −6 °C, and an average temperature of −11 °C. The temperature of exposed rock areas, including sunlit and shaded surfaces, shows a large difference between the maximum and minimum values, with the maximum reaching around 1 °C–2 °C and the minimum around −25 °C, resulting in an average temperature of and the minimum around −25 °C, resulting in an average −11.7 °C. Snow surface temperature predominantly ranges from and the minimum around −25 °C, resulting in an average of −23 °C to −13 °C, with an average temperature of −16.6 °C.

Fig. 10 illustrates the albedo of ice sheet surface features during mid-to-late January 202 derived from the MCD43A3 product. The albedo of wind-induced BIA and melt-induced BIA is quite close, predominantly ranging from 0.55 to 0.75. Among them, the average albedo of wind-induced BIA is slightly higher (0.67) than that of melt-induced BIA (0.63). The albedo of exposed rock areas exhibits a wide range of values (0.3–0.7) with an average albedo of 0.56. Within the "other" category, the proportion of supraglacial lakes relative to snow is negligible. Therefore, despite the low albedo of water, the average albedo of the "other" category remains relatively high at 0.79, with values generally ranging from 0.78 to 0.8. This is expected considering the dominant presence of snow within this category.

V. Discussion

The formation of blue ice is influenced by multiple factors, making it nearly impossible to achieve a completely accurate classification. In this study, the main source of uncertainty arises from the disregard of short-term significant changes in wind-induced BIAs caused by snowfall and the dynamics of the edge of wind-induced BIAs. While the accumulated snow above the wind-induced blue ice is removed at a faster rate due to the smooth surface and strong katabatic winds, snowfall can still induce temporary changes in these areas [2], [3], [6]. Nevertheless, the aforementioned phenomenon is limited to a small area, and the majority of wind-eroded BIAs remain highly stable. According to the accuracy verification of the classification results, it can be concluded that coherence analysis is an effective approach for blue ice classification in the LGB. We acknowledge that the distribution of blue ice obtained through properties such as snow grain size and albedo has a more solid physical foundation. However, the resolution of the currently available snow grain size data products is relatively low, and it is challenging to obtain high-precision snow grain size data through inversion. Albedo is influenced by factors such as bare rock, shadows, and surface melting. In addition, although theoretically feasible, utilizing SAR imagery for mapping BIA presents challenges, as discussed in the introduction. The radar backscatter is always noisy and susceptible to interference from various factors such as melting and radar penetration. Therefore, we did not incorporate SAR imagery into the blue ice mapping process. Instead, it was solely used for blue ice classification.

The results of blue ice classification based on coherence remain relatively unaffected by summer melting, as the presence of seasonal melting is one of the distinguishing factors between the two types of blue ice. This study also conducted blue ice classification experiments in the LGB using Sentinel-1 imagery from the frozen season and achieved outstanding classification results (Fig. 11). Coherence differences between the two types of blue ice were found in the nonmelting season, although they are less significant compared to summer. It indicates that the blue ice classification method introduced in this study showcases excellent robustness.

Due to the dynamic of BIAs, the extent of blue ice observed in this study may differ somewhat from previous studies [18]. We took deliberate measures to exclude supraglacial lakes that were mixed with blue ice, which was done by applying a near-infrared band threshold to differentiate them. It can

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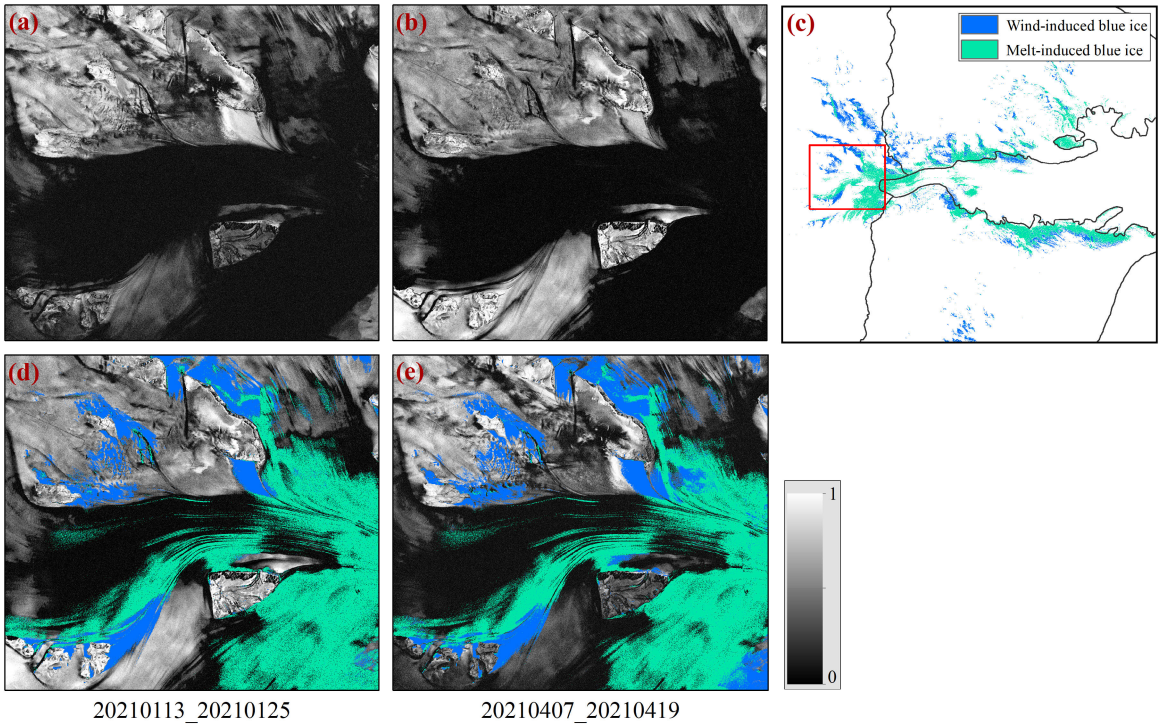


Fig. 11. Blue ice classification based on coherence images from the melting and nonmelting seasons in the LGB. (a) and (b) Coherence images for January and April. (c) Extent and location of the experimental area (the red box). (d) and (e) Blue ice classification results for the corresponding time.

minimize the potential influence on the analysis of atmospheric characteristics of different BIAs. In contrast, previous studies might not have explicitly distinguished supraglacial lakes from BIAs. As a result, the total estimated extent of BIAs in our study is slightly lower compared to the findings of previous studies. Specifically, the maximum BIAs extent acquired during the period from December 1999 to January 2003 was reported to be 2.04×10^4 km^2 [23], while Wei et al. [20] extracted a BIA of 2.47×10^4 km^2 based on Landsat 8 images from 2017 to 2019. In our study, the total extracted BIA amounted to 1.99×10^4 km^2, exhibiting a spatial distribution broadly consistent with prior investigations.

As described in Section IV-A, the melt-induced BIA, formed by surface melting and refreezing of ice layers, is mainly found downstream of glaciers and at the margins of the Amery ice shelf, as well as near Gillock Island, typically at altitudes below 1000 m (the average elevation is 437 m). Wind-induced BIA is typically found at relatively higher altitudes (the average elevation is 887 m), predominantly on leeward mountain peaks, glacier valleys, and upstream areas of glaciers [3]. In the cold and dry environment, the surface of ice sheet experiences minimal or no melting. The persistent strong katabatic winds caused erosion of the surface snow, resulting in the exposure of the underlying blue ice and preventing the accumulation of snow on the surface. The blue ice classification results in Lambert Glacier, Mellor Glacier, Fisher Glacier, and Mawson Escarpment are generally consistent with previous studies [23]. In contrast to the findings of Yu et al. [23], melt-induced BIA is found near Gillock Island, which is situated near the coastline of the ice sheet. This is in line with the findings of Hu et al. [6]. They reported that the average annual surface melting of the BIA near Gillock Island reached 87 mm, and the average daily 2-m temperature during the summer could reach as high as 3.3 °C between 2000 and 2021. These observations support the classification of the BIA in question as melt-induced, which aligns closely with our classification results.

Although the air temperature of melt-induced BIA is mostly negative, surface melting events can still occur during the summer [35]. It can be observed that supraglacial lakes and streams are frequently observed in the melt-induced BIAs during summer in optical satellite images. These liquid–water features are abundant at the onset of the ablation season and rapidly diminish when the season comes to an end. In contrast, surface water bodies are rarely found in wind-induced BIAs.

According to the statistical results presented in Section IV-B, the average surface albedo of melt-induced BIA and wind-induced BIA is 0.63 and 0.67, respectively, slightly higher than that estimated by Landsat 8 (ranging from 0.52 to 0.66) reported in the polar record glacier [16]. This is understandable, given that the lower resolution of the albedo product (MCD43A3) used in our study could have led to an overestimation of blue ice albedo due to the influence of surrounding snow surfaces. Overall, it can be found that the albedo of melt-induced BIA tends to be lower than that of wind-induced BIA. Although the repeated freeze-thaw cycles in melt-induced BIA increase its internal ice crystal content, thereby enhancing the albedo to some extent, the presence of meltwater still contributes to a slightly lower albedo compared to wind-induced BIA [2], [6], [36].

The statistical analysis of surface temperatures indicates that, in comparison to wind-induced BIAs (−11 °C), melt-induced BIAs exhibit higher average surface temperatures (−7.2 °C). This can be related to the distinct formation mechanisms of the two types of blue ice. Melt-induced blue ice is formed through seasonal melting and refreezing near the surface, and it naturally occurs in lower elevation coastal

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regions with higher surface temperatures [2], [6], [23]. In contrast, wind-induced blue ice originates from glacier ice formed by the accumulation of multiple layers of glaciers over time, which is typically found in dry and cold mountainous regions. Besides, the relatively lower albedo in ablation zones of blue ice may result in increased absorption of solar radiation, leading to higher surface temperatures.

Our classification of blue ice plays a crucial role in understanding the mass balance and surface energy budget of ice sheets. In the summer, the development of water systems is often observed in areas where melt-induced blue ice is located. These regions are influenced by surface melting, leading to a rapid transformation of the snow surface into slush, subsequently forming ponds or supraglacial lakes. In addition, the subsurface layers of the melt-induced BIAs contain a significant amount of subglacial meltwater and runoff [7], [8], [37], [38]. Due to the solid-state greenhouse effect [2], [8], shortwave radiation penetrates the surface layer of blue ice and causes subsurface melting within the ice [7], [11], [12], [13]. When surface and subsurface meltwater infiltrates the interior of the ice sheet, it results in dynamic changes in surface water systems and subglacial hydrological systems, thereby affecting the stability of the ice shelf [39]. In future studies, it is considered that the application of the method proposed in this article will be expanded to conduct blue ice classification on a larger scale. This would contribute to a deeper understanding of the hydrological processes and climate change of the Antarctic and provide a more comprehensive reference for airport site selection and meteorite search efforts.

VI. Conclusion

In this article, we utilize Sentinel-2 optical imagery from 2020 to 2022 with a 10-m resolution to map BIAs in the LGB of Antarctica and propose a new method to classify blue ice using dynamic coherence threshold based on Sentinel-1 SAR imagery. By conducting interferometric processing on a series of Sentinel-1 InSAR image pairs acquired in January 2021, coherent images were obtained. These images were then combined with visually determined discriminant samples of different types of blue ice to establish dynamic coherence thresholds suitable for distinguishing between melt- and wind-induced blue ice during that particular time period. The proposed method provided blue ice types with a spatial resolution of 10 m and demonstrated a high level of accuracy, achieving an overall accuracy of 0.909 and an $F1$-score of 0.906. In addition, statistical analysis was performed using DEM, temperature, and albedo data to explore the geographical characteristics and property differences of these two types of blue ice. The total area of the BIAs in the LGB obtained in this study is 1.986×10^4 km^2, with the area of melt-induced BIA being 1.276×10^4 km^2 and wind-induced BIA being 0.710×10^4 km^2. Our findings indicate that melt-induced BIA is primarily distributed in areas below 1000 m, while wind-induced BIA is prevalent across the range of 0–2000 m. Melt-induced BIA exhibits lower albedo and higher surface temperatures compared to wind-induced blue ice. Our methodology can be applied to large-scale and long-term studies related to blue ice. In the future, we will focus on investigating the seasonal and interannual variations of blue ice and exploring the underlying causes behind these changes.

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